FLIGHT MECHANICS DATA FOR WINDEX 1200 C

1. GEOMETRICAL REFERENCE DATA

1.1. WING AND AILERONS

b	SPAN OF WING	12.10 m
$\bar{c}$	MAC OF WING	0.633 m
$\bar{c}_A$	MAC OF AILERON	0.114 m
S	AREA OF WING	7.41 m^2
S_A	AREA OF AILERON (ONE AILERON)	0.266 m^2
E_A	CHORD RATIO OF AILERON	0.225
γ	DIHEDRAL ANGLE	3.0°

1.2. STABILIZER AND ELEVATOR

b_S	SPAN OF STABILIZER	2.18 m
$\bar{c}_S$	MAC OF STABILIZER	0.416 m
$\bar{c}_E$	MAC OF ELEVATOR	0.125 m
S_S	AREA OF STABILIZER	0.870 m^2
S_E	AREA OF ELEVATOR	0.261 m^2
E_E	CHORD RATIO OF ELEVATOR	0.30

1.3. FIN AND RUDDER

b_F	SPAN OF FIN	1.05 m
$\bar{c}_F$	MAC OF FIN	0.954 m
$\bar{c}_R$	MAC OF RUDDER	0.267 m
S_F	AREA OF FIN	0.95 m^2
S_R	AREA OF RUDDER	0.266 m^2
E_R	CHORD RATIO OF RUDDER	0.28

1.4. FUSELAGE AND TAIL LENGTHS

a_{FW}	MAX WIDTH OF FUSELAGE	0.60 m
a_{FH}	MAX HEIGTH OF FUSELAGE	0.75 m
a_{FL}	EFFERTIVE LENGTH OF FUSELAGE VOLUME	3.40 m
l_{tS}	DISTANCE BETWEEN $AC_{WING+FUS.}$ AND AC_{STAB}	2.643 m
l_{tF}	DISTANCE BETWEEN $AC_{WING+FUS.}$ AND AC_{FIN}	2.00 m

1.5. MASS AND MOMENT OF INERTIA DATA

M	MASS OF AIRCRAFT AT NORMAL WEIGHT	260 KG
I_{XX}	MOMENT OF INERTIA AROUND ROLL AXIS	600 $kg\,m^2$
I_{YY}	——— " ——— PITCH AXIS	240 $kg\,m^2$
I_{ZZ}	——— " ——— YAW AXIS	780 $kg\,m^2$

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2. DEFINITIONS

2.1. LENGTHS

b	WING SPAN	[m]
c	CHORD	"
$\bar{c}$	WING MAC	"
l_t	TAIL LENGTH	"
x_{SM}	STABILITY MARGIN	"

x	CHORDWISE COORDINATE	
y	SPANWISE —— " ——	
z	VERTICAL —— " ——	

2.2. AREA

S	WING AREA	[m^2]

2.3. VELOCITIES

V	FLIGHT SPEED	[m/s]
p	ROLL ANGULAR VELOCITIES	[RAD/s]
q	PITCH —— " ——	"
r	YAW —— " ——	"

2.4. ANGLES

α	ANGLE OF ATTACK	[RAD]
β	—— " —— YAW	"
δ	DEFLECTION OF CONTROL SURFACE	"
γ	ANGLE OF DIHEDRAL	"
$\bar{p} = p \cdot b / 2V$	HELICAL TIP PATH ROLL ANGLE	"
$\bar{q} = q \cdot \bar{c} / V$	FLIGT PATH ANGLE OF CURVATURE IN PITCH	"
$\bar{r} = r \cdot b / 2V$	—— " —— YAW	"
ε	ANGLE OF DOWN WASH	

2.5. FORCES

L	LIFT	[N]
D	DRAG	"
C	SIDE FORCE	"

2.6. MOMENTS

l	ROLLING MOMENT	[Nm]
m	PITCHING — " —	"
n	YAWING — " —	"
h	HINGE — " —	"

2.7 SUBSCRIPTS

A	AILERON
E	ELEVATOR
F	FIN
R	RUDDER
S	STABILIZER
FIX	STICK FIXED
FREE	STICK FREE
∞	IN INFINITY

2.8. PRESSURE

$\frac{1}{2}\rho V^2$ DYNAMIC PRESSURE [N/m²]

ρ DENSITY, AT S.L. $\rho \simeq 1.226$ [kg/m³]

2.9 AERODYNAMIC COEFFICIENTS

2.9.1. LONGITUDINAL

$C_D = D/\frac{1}{2}\rho V^2 \cdot S$ DRAG COEFFICIENT

$C_L = L/\frac{1}{2}\rho V^2 \cdot S$ LIFT —— " ——

$C_m = m/\frac{1}{2}\rho V^2 S \cdot \bar{c}$ PITCHING MOMENT —— " ——

$C_{hE} = h_E/\frac{1}{2}\rho V^2 S_E \cdot \bar{c}_E$ ELEVATOR HINGE MOMENT —— " ——

2.9.2. LATERAL

$C_C = C/\frac{1}{2}\rho V^2 S$ SIDE FORCE COEFFICIENT

$C_l = l/\frac{1}{2}\rho V^2 S\, b/2$ ROLLING MOMENT —— " ——

$C_n = n/\frac{1}{2}\rho V^2 S\, b/2$ YAWING MOMENT —— " ——

C_{hA} $h_A/\frac{1}{2}\rho V^2 S_A \cdot \bar{c}_A$ AILERON HINGE MOMENT —— " —— (PER AILERON)

C_{hR} $h_R/\frac{1}{2}\rho V^2 S_R\, \bar{c}_R$ RUDDER —— " —— —— " ——

2.10. TABLES OF AERODYNAMIC STABILITY DERIVATIVE

A. LONGITUDINAL

	α	$\bar{q}$	δ_E
C_L	$C_{L\alpha}$	C_{Lq}	$C_{L\delta E}$
C_m	$C_{m\alpha}$	$C_{m\bar{q}}$	$C_{m\delta E}$
C_{hE}	$C_{hE\alpha}$	$C_{hE\bar{q}}$	$C_{hE\delta E}$

B. LATERAL

	β	$\bar{p}$	$\bar{r}$	δ_A	δ_R
C_C	$C_{C\beta}$	——	$C_{C\bar{r}}$	——	$C_{C\delta R}$
C_l	$C_{l\beta}$	$C_{l\bar{p}}$	$C_{l\bar{r}}$	$C_{l\delta A}$	——
C_n	$C_{n\beta}$	$C_{n\bar{p}}$	$C_{n\bar{r}}$	——	$C_{n\delta R}$
C_{hA}	——	$C_{hA\bar{p}}$	——	$C_{hA\delta A}$	——
C_{hR}	$C_{hR\beta}$	——	$C_{hR\bar{r}}$	——	$C_{hR\delta R}$

3. AERODYNAMIC DATA

3.1 TWO-DIMENSIONAL SECTION DATA (FROM DATA IN [1]

$C_{l\alpha} = 6.59$ [1/RAD] $\quad AC_\alpha$ ON 26.0 % CHORD

EFFECTIVE ANGLE OF ATTACK IN CURVED FLOW IS AT 75% CHORD

E	0.225	0.280	0.300
$C_{l\delta}/C_{l\alpha}$	0.510	0.575	0.600
$C_{h\alpha}$	-0.250	-0.285	-0.300
$C_{h\delta}$	-0.595	-0.598	-0.600
$\frac{d\delta}{d\alpha}(C_h=0)$	-0.420	-0.477	-0.500
$C_{l\alpha}$ FREE / $C_{l\alpha}$ FIX	0.786	0.726	0.700
AC_δ % C	43.3	41.4	41.0

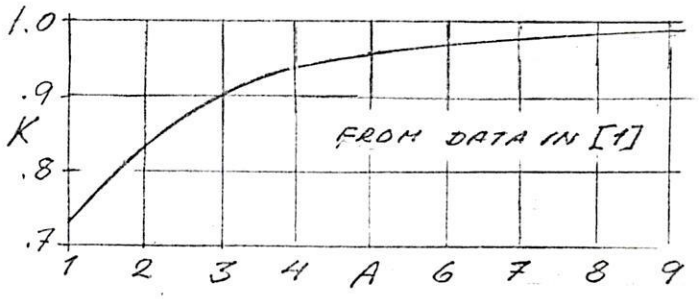
EFFECTIVE CHORD REDUCTION DUE TO FREE FLOATING CONTROL SURFACE

3.2 THREE-DIMENSIONAL EFFECTS

$A = b^2/S$ ASPECT RATIO

$$C_{L\alpha} = K \frac{A}{A+2} C_{l\alpha} = \eta(A) C_{l\alpha}$$

K(A) AS SHOWN IN DIAGRAM.



LIFT, PITCHING MOMENT AND HINGE MOMENT α-AND δ-DERIVATIVES ARE ASSUMED TO DEPEND ON A AS $C_{L\alpha}$.

3.3 THREE-DIMENSIONAL DATA

WING: $A = 12.10^2/7.41 = 19.76 \quad K \approx 1.00 \quad \eta_W = 0.9081$

STAB S. FIX: $A = 2.18^2/0.87 = 5.46 \quad K = 0.96 \quad \eta_{S.FIX} = 0.7026$

STAB S. FREE: $A = 0.7 \cdot 5.46 = 7.80 \quad K = 0.98 \quad \eta_{S.FREE} = 0.7800 \quad (S_{S_{EFF}} = 0.7 \cdot S_S)$

FIN: $A = 1.05^2/0.95 = 1.16$ EFFECT OF STABILIZER IS APPROXIMATELY TO INCREASE EFFECTIVE A BY 50% [1]
$A_{EFF} = 1.5 \cdot 1.19 = 1.74 \quad K = 0.82 \quad \eta_F = 0.3815$

FOR CALCULATION OF LATERAL STABILITY DERIVATES WHERE THE WING FLOW IS BASICALLY ANTISYMMETRICAL ONLY HALF THE NOMINAL VALUE OF THE ASPECT RATIO IS ASSUMED:

η_W (LAT) = 0.8233

4. CALCULATIONS OF LONGITUDINAL AERODYNAMIC DATA

4.1. WING ONLY

$C_{L\alpha W} = 0.9081 \cdot 6.59 = 5.984$ [1/RAD]

AC ON 26% $\bar{C}$ EQUAL TO 214.0 mm BEHIND WING ROOT LEADING EDGE, SEE FIG 2

4.2. EFFECT OF FUSELAGE

THE SCHEMATIC EFFECT OF FUSELAGE FROM α IS A PURE PITCHING MOMENT THAT MOVES THE AC FORWARDS. THE FUSELAGE IN PITCH IS APPROXIMATED BY A PARABOLA OF ROTATION WITH A MAX DIAMETER OF 0.60 m AND A LENGTH OF 3.40 m.

THE VOLUME IS $VOL_{FUS} = 8/15 \cdot \pi/4 \cdot 0.60^2 \cdot 3.4 = 0.51\, m^3$

THE PITCHING MOMENT $m_{FUS} = 1/2\, \rho V^2 \alpha\, k\, VOL_{FUS}$

$d/l = 0.6/3.4 = 0.18 \qquad k = 2 \cdot 0.85 = 1.7$ [1]

$\Delta X_{AC}(FUS) = m_{FUS} / 1/2\, \rho V^2 S_W \alpha\, C_{L\alpha W} = \dfrac{1.7 \cdot 0.51}{7.41 \cdot 5.984} = 0.020\, m$

AC (WING + FUS.) 214 − 20 = 194 BEHIND WING ROOT L.E.

4.3. EFFECT OF STABILIZER

4.3.1. DOWNWASH FROM WING

THE FAR FIELD DOWN WASH ANGLE IN THE WAKE PLANE BEHIND AN ELLIPTICALLY LOADED WING IS:

$\varepsilon_\infty = 2\, C_L / \pi A$ CLOSER TO THE WING THE ANGLE IS STEEPER

WITH $2X_S/b = 0.433$ REF [1] GIVES $\varepsilon = 2.85\, C_L/\pi A$

WITH THE T-TAIL CONFIG. WITH THE STABILIZER SET ABOVE THE PLANE OF THE WING THE RELATIVE VERTICAL DISTANCE

$2 Z_S/b \approx 0.1$ GIVES A DOWNWASH REDUCTION OF 6%

THE DOWNWASH ANGLE FOR THE STABILIZER $\varepsilon_S = 0.94 \cdot 2.85 \dfrac{C_{LW}}{\pi A}$

THE EFFECTIV α OF THE STABILIZER $\alpha_S = \alpha_\infty - \varepsilon_S$:

$\alpha_S = \alpha_\infty (1 - 0.94 \cdot 2.85\, C_{L\alpha W} / \pi A) = 0.742\, \alpha_\infty$

$1 - \dfrac{\partial \varepsilon}{\partial \alpha} = 0.742$

4.3.2. LIFT OF COMPLETE CONFIGURATION, STICK FIX.

$$C_{L\alpha\ WING+FUS+STAB,FIX} = C_{L\alpha\ WING} + \frac{S_s}{S} \cdot \eta_{SFIX} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) c_{l\alpha}$$

$$C_{L\alpha}\ \text{—}\,''\,\text{—} = 5.984 + \frac{0.870}{7.41} \cdot 0.7026 \cdot 0.742 \cdot 6.59$$

$$C_{L\alpha}\ \text{—}\,''\,\text{—} = 5.984 + 0.4034 = 6.387$$

4.3.3. LIFT OF COMPLETE CONFIGURATION, STICK FREE

$$C_{L\alpha\ WING+FUS+STAB,FREE} = 5.984 + \frac{0.7 \cdot 0.87}{7.41} \cdot 0.7800 \cdot 0.742 \cdot 6.59$$

$$C_{L\alpha}\ \text{—}\,''\,\text{—} = 5.984 + 0.3135 = 6.297$$

4.5 CALCULATION OF DERIVATIVES DUE TO PITCH ROTATION

4.5.1. $C_{L\bar{q}}$

ANGULAR VELOCITY q AROUND C.G

CONTRIBUTION FROM WING, EFFECTIVE AOA AT 75% $\bar{C}_W$

$$L_W = 1/2 \rho V^2 \cdot S \cdot \eta_W \cdot C_{l\alpha} \cdot q \cdot (X_{SM} + 0.163)/V$$

CONTRIBUTION FROM STABILIZER, EFFECTIVE AOA AT 75% $\bar{C}_S$

$$L_S = 1/2 \rho V^2 \cdot S_S \cdot \eta_S \, C_{l\alpha} \cdot q \cdot (X_{SM} + 2.680)/V$$

$$C_{L\bar{q}} = C_L / \frac{q\bar{c}}{V}$$

$$C_{L\bar{q}W} = \eta_W \cdot C_{l\alpha} \frac{X_{SM} + 0.163}{\bar{c}} = 0.9081 \cdot 6.59/0.633 \, (X_{SM} + 0.163)$$

$$C_{Lq_S} = \frac{S_S}{S} \cdot \eta_S \, C_{l\alpha} \frac{X_{SM} + 2.680}{\bar{c}} = \frac{0.87}{7.41} \cdot 0.7026 \cdot 6.59/0.633 \, (X_{SM} + 2.680)$$

$$C_{L\bar{q}W} = 9.454 \, (X_{SM} + 0.163) = 9.454 \, X_{SM} + 1.541$$

$$C_{L\bar{q}S} = 0.8588 \, (X_{SM} + 2.680) \qquad 0.859 \, X_{SM} + 2.302$$

$$C_{L\bar{q}W+S} = C_{L\bar{q}} = 10.31 \, X_{SM} + 3.843$$

4.5.2. $C_{m\bar{q}}$

EFFECTIVE C.E OF PANEL LIFT AS ESTIMATED BELOW

CONTRIBUTION FROM WING: C.E AT ABOUT 40% $\bar{C}_W$

$$C_{m\bar{q}W} = -(9.454 \, X_{SM} + 1.541) \frac{X_{SM} - 0.058}{\bar{c}}$$

CONTRIBUTION FROM STABILIZER: C.E. AT ABOUT 27% $\bar{C}_S$

$$C_{m\bar{q}S} = -(0.859 \, X_{SM} + 2.302) \frac{X_{SM} + 2.480}{\bar{c}}$$

$$-C_{m\bar{q}W} = (9.454 X_{SM} + 1.541)(1.580 X_{SM} + 0.092) = 14.94 X_{SM}^2 + 3.305 X_{SM} + 0.14$$

$$-C_{m\bar{q}S} = (0.859 X_{SM} + 2.302)(1.580 X_{SM} + 3.918) = 1.36 X_{SM}^2 + 7.003 X_{SM} + 9.01$$

$$-C_{m\bar{q}W+S} = 16.30 X_{SM}^2 + 10.308 X_{SM} + 9.161$$

$$X_{SM} \approx 0.1 \, m \qquad 16.30 X_{SM}^2 \approx 1.63 X_{SM}$$

$$C_{m\bar{q}} \approx -11.94 X_{SM} - 9.161$$

4.5.3. $C_{h_E \bar{q}}$

EFFECTIVE AOA IS ASSUMED TO BE AT HINGE LINE, 70% $\bar{c}$

$$\alpha_{EFF} = q \cdot (X_{SM} + 2.659)/V = \frac{q\bar{c}}{V} \frac{(X_{SM} + 2.659)}{\bar{c}}$$

$$\alpha_{EFF}/\bar{q} = \frac{X_{SM} + 2.659}{\bar{c}}$$

$$C_{h_E \bar{q}} = \eta_S \cdot C_{h_\alpha} \cdot \alpha_{EFF}/\bar{q} = C_{h_{E\alpha}} \frac{X_{SM} + 2.659}{\bar{c}} =$$

$$C_{h_E \bar{q}} = 0.7026(-0.300) \frac{X_{SM} + 2.659}{0.633} = -0.333\, X_{SM} - 0.8854$$

NOTE: S_{SE} AND $\bar{c}_{SE}$ BEING REF VALUES FOR C_{h_E}

4.6. REMAINING LONGITUDINAL DERIVATIVES

4.6.1. C_{m_α}

$$C_{m_\alpha} = -C_{L_\alpha} \cdot \frac{X_{SM}}{\bar{c}} = -\frac{6.387}{0.633} \cdot X_{SM} = -10.09\, X_{SM}$$

4.6.2 $C_{h_{E\alpha}}$

$$C_{h_{E\alpha}} = \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) \eta_S \cdot C_{h_\alpha} = 0.742 \cdot 0.7026(-0.300) = -0.156$$

4.6.3. $C_{L_{\delta_E}}$

$$C_{L_{\delta_E}} = \frac{S_S}{S} \cdot \eta_S\, C_{L_\alpha} \cdot C_{L_\delta}/C_{L_\alpha} = \frac{0.87}{7.41} \cdot 0.7026 \cdot 6.59 \cdot 0.600 = 0.326$$

4.6.4. $C_{m_{\delta_E}}$

$$C_{m_{\delta_E}} = -C_{L_{\delta_E}} \cdot \frac{X_{SM} + l_{tS}(\delta_E)}{\bar{c}} = -0.326 \cdot \frac{X_{SM} + 2.538}{0.633} = -0.515\, X_{SM} - 1.308$$

4.6.5. $C_{h_{E\delta_E}}$

$$C_{h_{E\delta_E}} = \eta_S \cdot C_{h_{E\delta_E}} = 0.7026 \cdot (-0.600) = -0.422$$

5. SUMMARY OF LONGITUDINAL STABILITY DERIVATIVES

5.1. MORE COMPLETE EXPRESSIONS

	α	$\bar{q}$	δ_E
C_L	6.387	$10.31\,X_{SM} + 3.843$	0.326
C_m	$-10.09\,X_{SM}$	$-11.94\,X_{SM} - 9.161$	$-0.515\,X_{SM} - 1.308$
C_{hE}	-0.156	$-0.333\,X_{SM} - 0.885$	-0.422

5.2. SIMPLYFIED EXPRESSIONS (ASSUMING $X_{SM} = 0.10$)

	α	$\bar{q}$	δ_E
C_L	6.387	4.874	0.326
C_m	$-10.09\,X_{SM}$	-10.355	-1.360
C_{hE}	-0.156	-0.918	-0.422

6. CALCULATIONS OF LATERAL AERODYNAMIC DATA

6.1. C_{C_β}

CONTRIBUTION FROM FIN: $C_{C_{\beta FIN}} = \frac{S_F}{S} \cdot \eta_F \cdot C_{l\alpha} = \frac{0.95}{7.41} \, 0.3815 \cdot 6.59 = 0.3223$

CONTRIBUTION FROM WING $C_{C_{\beta WING}} = \gamma^2 \eta_W(LAT) \cdot C_{l\alpha} = \left(\frac{3}{57.3}\right)^2 \cdot 0.8233 \cdot 6.59 = 0.0149$

$$C_{C_{\beta F+W}} = C_{C_\beta} = 0.337$$

6.2. $C_{C_{\bar{r}}}$

ONLY CONTRIBUTION FROM FIN. EFFECTIVE AOA AT 75% $\bar{C}_F$

$\bar{r}$ - ROTATION AROUND C.G RADIUS $1.833 + \frac{\bar{C}_F}{2} + X_{SM} = 2.310 + X_{SM}$

$$C_{C_{\bar{r}}} = -C_{C_{\beta FIN}} \cdot \frac{2.310 + X_{SM}}{b/2} = -0.3223 \, \frac{2.310 + X_{SM}}{6.05} = -0.053 \, X_{SM} - 0.123$$

6.3. $C_{C_{\delta R}}$

$$C_{C_{\delta R}} = C_{C_{\beta FIN}} \cdot C_{l\delta R}/C_{l\alpha} = 0.3223 \cdot 0.575 = 0.185$$

6.4 C_{l_β}

CONTRIBUTION FROM WING: SPANWISE C.E. OF PANEL LIFT IS AT $y = 2.833 \, m = y_{C.E.}$

$$-C_{l_{\beta WING}} = \gamma \cdot \eta_W(LAT) \cdot C_{l\alpha} \frac{2 \cdot y_{C.E.}}{b} = \frac{3}{57.3} \, 0.8233 \cdot 6.59 \, \frac{2 \cdot 2.833}{12.1} = 0.1330$$

CONTRIBUTION FROM T-TAIL: VERTICAL C.E OF FIN PANEL SIDE FORCE IS AT $z = -0.45 \, m = z_{C.E.}$

$$-C_{l_{\beta FIN}} = C_{C_{\beta FIN}} \cdot \frac{2 \cdot z_{C.E}}{b} = 0.3225 \cdot \frac{2 \cdot 0.45}{12.1} = 0.0240$$

THE FIN INDUCED LIFT ON EACH SEMI STABILIZER PANEL IS ASSUMED TO BE HALF THE FIN SIDE FORCE. C.E AT (STAB SEMI SPAN)/3

$$-C_{l_{\beta STAB}} = \frac{C_{C_{\beta FIN}}}{2} \cdot \frac{b_s \cdot 2}{6 b} \cdot 2 = 0.3225 \cdot 0.060 = 0.0194$$

$$C_{l_{\beta WING + T\text{-}TAIL}} = C_{l_\beta} = -0.176$$

6.5. $C_{l_{\bar{p}}}$

APPROXIMATING THE WING PLANFORM BY AN ELLIPSE AND NEGLECTING CONTRIBUTIONS FROM THE TAIL $C_{l_{\bar{p}}}$ IS GIVEN BY:

$$C_{l_{\bar{p}}} \approx -\frac{C_{L\alpha W}}{4} = -\frac{\eta_W(LAT) \cdot C_{l\alpha}}{4} = -\frac{0.8233 \cdot 6.59}{4} = -1.356$$

6.6. $C_{l\bar{r}}$

ASSUMING AN ELLIPTICAL SPANWISE LIFT DISTRIBUTION ON THE WING IN STRAIGHT (NON CURVING) FLIGHT AND NEGLECTING TAIL CONTRIBUTIONS:

$$C_{l\bar{r}} = \frac{C_{L_W}}{2} = C_L \cdot \frac{C_{L_W}}{C_L} \cdot 0.5 \approx 0.5\, C_L$$

6.7. $C_{l\delta_A}$

MOMENTARM FOR AILERONS $y_A = 4.609\ m$ AREA OF AILERON COVERED WING AREA $1.225\ m^2$ (EACH WING PANEL) SPAN OF AILERON COVERED WING AREA $= 2.50\ m$, CORRESPONDING ASPECT RATIO $= 5.10$

$\eta_A = 0.6896 \qquad E = 0.225 \qquad C_{l\delta}/C_{l\alpha} = 0.510$

$$C_{l\delta_A} = 2 \cdot \frac{S_A}{S}\, \eta_A \cdot \frac{C_{l\delta}}{C_{l\alpha}} \cdot C_{l\alpha} \frac{2 \cdot y_A}{b} = 2\, \frac{1.225}{7.41} \cdot 0.6896 \cdot 0.51 \cdot 6.59 \cdot \frac{2 \cdot 4.609}{12.1} = 0.584$$

6.8. $C_{n\beta}$

CONTRIBUTION FROM FIN; TAIL LENGT $l_{tF} = 1.833\ m$

$$C_{n\beta\,FIN} = C_{C\beta\,FIN} \frac{2(l_{tF} + X_{SM})}{b} = 0.3223\, \frac{2(1.833 + X_{SM})}{12.1} = 0.0976 + 0.053\, X_{SM}$$

CONTRIBUTION FROM FUSELAGE. THE HEIGHT OF THE FUSELAGE IS 50% MORE THAN THE WIDTH. EFFECTIVE VOLUME IN YAW IS THUS $1.5 \cdot 0.51 = 0.77\ m^3$ (SEE 4.2). $d/l = 1.5 \cdot 0.18 = 0.27$ GIVING $K = 1.48$

$$C_{n\beta\,FUS} = -\frac{K\,(\text{EFF. VOLUME})}{S \cdot b/2} = -\frac{2 \cdot 1.48 \cdot 0.77}{7.41 \cdot 12.1} = -0.0254$$

THE EFFECT OF WING ON YAWING MOMENT IS SMALL AND NEGLECTED.

$$C_{n\beta\,FIN+FUS.} = C_{n\beta} = 0.0976 + 0.053\, X_{SM} - 0.0254 = 0.0722 + 0.053\, X_{SM}$$

6.9. $C_{n\bar{p}}$

CONTRIBUTION FROM WING ASSUMING ELLIPTICAL PLANFORM

$$C_{n\bar{p}\,WING} = -C_{L_{WING}}/4 \approx -0.25\, C_L \quad \text{(ADVERSE YAW)}$$

CONTRIBUTION FROM FIN, C.E. FOR FIN SIDE FORCE $Z \approx = -0.6\ m$

$$C_{n\bar{p}\,FIN} = \frac{(-Z)}{b/2} \cdot \frac{S_E}{S} \cdot \eta_F \cdot C_{l\alpha} = \frac{2 \cdot 0.6 \cdot 0.95}{12.1 \cdot 7.41}\, 0.3815 \cdot 6.59 = 0.032$$

$$C_{n\bar{p}\,WING+FIN} = C_{n\bar{p}} = -0.25\, C_L + 0.032$$

6.10. $C_{n\bar{r}}$

ONLY CONTRIBUTIONS FROM THE FIN
THE C.E. OF THE FIN SIDE FORCE IS ESTIMATED TO BE AT 31% $\bar{c}_F$
MOMENT ARM $1.833 + 0.05 \cdot \bar{c}_F + X_{SM} = 1.833 + 0.05 \cdot 0.954 + X_{SM} = 1.881 + X_{SM}$

$$C_{n\bar{r}_{FIN}} = C_{n\bar{r}} = C_{C\bar{r}} \cdot \frac{2(1.881 + X_{SM})}{b} = (-0.053 X_{SM} - 0.123)(0.311 + 0.165 X_{S}$$

$$C_{n\bar{r}} = -(0.0383 + 0.0368 X_{SM} + 0.0087 X_{SM}^2) \approx -(0.0383 + 0.0377 X_{SM})$$

6.11. $C_{n\delta R}$

THE C.E OF THE RUDDER INDUCED SIDE FORCE ON THE FIN IS AT 41.4% $\bar{c}_F$.
MOMENT ARM $= 1.833 + 0.154 \cdot 0.954 + X_{SM} = 1.980 + X_{SM}$

$$C_{n\delta R_{FIN}} = C_{n\delta R} = C_{C\delta R} \cdot \frac{1.980 + X_{SM}}{b/2} = 0.185 \frac{2(1.980 + X_{SM})}{12.1}$$

$$C_{n\delta R} = 0.0605 + 0.031 X_{SM}$$

6.12. $C_{hA\bar{p}}$

$E = 0.225$ FOR THE AILERONS. EFFECTIVE RADIUS FOR p-ROTATION
$y_{EFF} = 4.7$ m

$$C_{hA\bar{p}} = C_{h\alpha A} \cdot \eta_W(LAT) \cdot \frac{2 \cdot y_{EFF}}{b} = -0.250 \cdot 0.8233 \cdot \frac{2 \cdot 4.7}{12.1} = -0.160$$

6.13. $C_{hA\delta A}$

$$C_{hA\delta A} = C_{h\delta A} \cdot \eta_A = +0.595 \cdot 0.6896 = +0.410 \quad \text{(SEE 6.7.)}$$

6.14. $C_{hR\beta}$

$$C_{hR\beta} = C_{h\alpha} \cdot \eta_F = -0.285 \cdot 0.3815 = -0.109$$

6.15. $C_{hR\bar{r}}$

EFFECTIVE AOA IS ASSUMED TO BE ON THE HINGE LINE 72% $\bar{c}_F$
EFFECTIVE RADIUS $1.833 + 0.46 \cdot 0.954 + X_{SM} = 2.272 + X_{SM}$

$$C_{hR\bar{r}} = -C_{hR\beta} \cdot \frac{2(2.272 + X_{SM})}{b} = 0.109 \frac{2(2.272 + X_{SM})}{12.1} = 0.0409 + 0.018 X_{S}$$

6.16. $C_{hR\delta R}$

$$C_{hR\delta R} = C_{hR\beta} \cdot \frac{C_{hR\delta R}}{C_{hR\alpha}} = -0.109 \cdot \frac{-0.598}{-0.285} = -0.229$$

7. SUMMARY OF LATERAL STABILITY DERIVATIVES

7.1. MORE COMPLETE EXPRESSIONS

	β	$\bar{p}$	$\bar{r}$	δ_A	δ_R
C_C	0.0337	—	$-0.053 X_{SM} - 0.123$	—	0.185
C_l	-0.176	-1.356	$0.5 C_L$	0.584	—
C_n	$0.053 X_{SM} + 0.0722$	$-0.25 C_L + 0.032$	$-0.038 X_{SM} - 0.383$	—	$0.031 X_{SM} + 0.0605$
C_{hA}	—	-0.160	—	0.410	—
C_{hR}	-0.109	—	$0.018 X_{SM} + 0.0409$	—	-0.229

7.2. SIMLYFIED EXPRESSIONS (ASSUMING $X_{SM} = 0.10$)

	β	$\bar{p}$	$\bar{r}$	δ_A	δ_R
C_C	0.0337	—	-0.128	—	0.185
C_l	-0.176	-1.356	$0.5 C_L$	0.584	—
C_n	0.0775	$-0.25 C_L + 0.032$	-0.387	—	0.0636
C_{hA}	—	-0.160	—	0.410	—
C_{hR}	-0.109	—	0.0427	—	-0.229